

RWorksheet_prado#3a

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```
LETTERS[1:11]

## [1] "A" "B" "C" "D" "E" "F" "G" "H" "I" "J" "K"
LETTERS[seq(1, 25, by = 2)]

## [1] "A" "C" "E" "G" "I" "K" "M" "O" "Q" "S" "U" "W" "Y"
LETTERS[c(1, 5, 9, 15, 21)]

## [1] "A" "E" "I" "O" "U"
letters[22:26]

## [1] "v" "w" "x" "y" "z"
tail(letters, 5)

## [1] "v" "w" "x" "y" "z"
letters[15:24]

## [1] "o" "p" "q" "r" "s" "t" "u" "v" "w" "x"
city <- c("Tuguegarao City", "Manila", "Iloilo City", "Tacloban", "Samal Island", "Davao City")
print(city)

## [1] "Tuguegarao City" "Manila"          "Iloilo City"      "Tacloban"
## [5] "Samal Island"    "Davao City"
temp <- c(42, 39, 34, 34, 30, 27)
print(temp)

## [1] 42 39 34 34 30 27
april_temp_df <- data.frame(city, temp)
print(april_temp_df)

##           city temp
## 1 Tuguegarao City  42
## 2      Manila    39
## 3   Iloilo City   34
## 4    Tacloban    34
## 5  Samal Island   30
## 6    Davao City   27
names(april_temp_df) <- c("City", "Temperature")
print(april_temp_df)

##           City Temperature
```

```

## 1 Tuguegarao City      42
## 2      Manila         39
## 3      Iloilo City     34
## 4      Tacloban       34
## 5      Samal Island    30
## 6      Davao City      27

str(april_temp_df)

## 'data.frame':  6 obs. of  2 variables:
## $ City      : chr  "Tuguegarao City" "Manila" "Iloilo City" "Tacloban" ...
## $ Temperature: num  42 39 34 34 30 27

april_temp_df[3:4, ]

##           City Temperature
## 3 Iloilo City           34
## 4  Tacloban            34

highest_temp_city <- april_temp_df[april_temp_df$Temperature == max(april_temp_df$Temperature), ]
print("City with Highest Temperature:")

## [1] "City with Highest Temperature:"
print(highest_temp_city)

##           City Temperature
## 1 Tuguegarao City      42

lowest_temp_city <- april_temp_df[april_temp_df$Temperature == min(april_temp_df$Temperature), ]
print("City with Lowest Temperature:")

## [1] "City with Lowest Temperature:"
print(lowest_temp_city)

##           City Temperature
## 6 Davao City           27

data_vector <- c(1:8, 11:14)
original_matrix <- matrix(data_vector, nrow = 3, ncol = 4)
original_matrix

##           [,1] [,2] [,3] [,4]
## [1,]      1      4      7     12
## [2,]      2      5      8     13
## [3,]      3      6     11     14

matrix_b <- original_matrix * 2
matrix_b

##           [,1] [,2] [,3] [,4]
## [1,]      2      8     14     24
## [2,]      4     10     16     26
## [3,]      6     12     22     28

original_matrix[2, ]

## [1]  2  5  8 13

```

```

original_matrix[1:2, 3:4]

##      [,1] [,2]
## [1,]    7  12
## [2,]    8  13

# Alternative: original_matrix[c(1, 2), c(3, 4)]
original_matrix[3, c(2, 3)]

## [1]  6 11

original_matrix[, 4]

## [1] 12 13 14

rownames(matrix_b) <- c("isa", "dalawa", "tatlo")
colnames(matrix_b) <- c("uno", "dos", "tres", "quatro")
matrix_b

##      uno dos tres quatro
## isa      2  8  14    24
## dalawa   4 10  16    26
## tatlo    6 12  22    28

dim(original_matrix) <- c(6, 2)
original_matrix

##      [,1] [,2]
## [1,]    1    7
## [2,]    2    8
## [3,]    3   11
## [4,]    4   12
## [5,]    5   13
## [6,]    6   14

data_vector <- c(1, 2, 3, 6, 7, 8, 9, 0, 3, 4, 5, 1)
repeated_data <- rep(data_vector, each = 2)
three_d_array <- array(repeated_data, dim = c(2, 4, 3))
three_d_array

## , , 1
##
##      [,1] [,2] [,3] [,4]
## [1,]    1    2    3    6
## [2,]    1    2    3    6
##
## , , 2
##
##      [,1] [,2] [,3] [,4]
## [1,]    7    8    9    0
## [2,]    7    8    9    0
##
## , , 3
##
##      [,1] [,2] [,3] [,4]
## [1,]    3    4    5    1
## [2,]    3    4    5    1

```

```
##"Your array has three dimensions."
```

```
dim_names <- list(  
  row_names = c("a", "b"),  
  col_names = c("A", "B", "C", "D"),  
  array_names = c("1st-Dimensional Array", "2nd-Dimensional Array", "3rd-Dimensional Array")  
)
```

```
named_array <- array(repeated_data, dim = c(2, 4, 3), dimnames = dim_names)  
named_array
```

```
## , , array_names = 1st-Dimensional Array
```

```
##
```

```
##      col_names
```

```
## row_names A B C D
```

```
##      a 1 2 3 6
```

```
##      b 1 2 3 6
```

```
##
```

```
## , , array_names = 2nd-Dimensional Array
```

```
##
```

```
##      col_names
```

```
## row_names A B C D
```

```
##      a 7 8 9 0
```

```
##      b 7 8 9 0
```

```
##
```

```
## , , array_names = 3rd-Dimensional Array
```

```
##
```

```
##      col_names
```

```
## row_names A B C D
```

```
##      a 3 4 5 1
```

```
##      b 3 4 5 1
```